Reward function

$$r_t = -\sum_{k=1}^{5} \left(w_1 x_{1,k} + w_2 x_{2,k} + w_3 x_{3,k} + w_4 \left(1 - \cos \psi_t\right) + w_5 \right) \tag{1}$$

where k runs over the 5 simulated seconds of decision step t, and:

- $x_{1,k}$ is the number of aircraft within 5 NM in second k (usually 0 or 1), with $w_1 = 100$;
- $x_{2,k} \in \{0, 1\}$ is 1 if the aircraft is inside a restricted area in second k, with $w_2 = 10$;
- $x_{3,k} \in \{0, 1\}$ is 1 if the aircraft is outside the sector in second k, with $w_3 = 10$;
- ψ_t is the angle between the aircraft's heading and the bearing to its waypoint at the end of the step, with $w_4 = 1$;
- $w_5 = 0.1$ is a time penalty per second.

So five seconds in a loss of separation cost 500 and five seconds in a restricted area 50. The term $1 - \cos \psi_t$ is 0 when flying straight at the waypoint and 2 when flying away from it. If the waypoint is reached during a step, the sum runs only over the seconds flown. The reward is never positive.